Pantry and Plate Video Demonstration Script

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Target duration 6 minutes 40 seconds. Record at 1080p with clear voice narration. Keep Android Studio, the running emulator and the repository commit page ready. Rehearse this guide and explain the code in your own words.

Before recording: clear the pantry through the interface. Prepare bread 2 piece, tomatoes 1 piece and olive oil 5 ml. Tomato toast requires these exact three ingredients. Use a future expiry date if entering one. Add the video file to the final ZIP yourself and keep the complete submission under 50 MB.

0:00 to 1:00 GitHub walkthrough

Say: This is Pantry and Plate, my Java Smart Pantry Manager. The repository contains the complete Android Studio project, README, tests and development history. Scroll from the earliest commit to the latest and describe the implementation order. Accurately disclose assistance and the actual development dates. Do not describe unpushed local commits as online history.

Commit order: The history begins with the Java project and README, then adds the models and matcher, SQLite CRUD, recipe seeds, shared UI, pantry adapter, validated editor, recipe suggestions, recipe detail and settings. Later commits add regression tests, Android 15 insets, the Gradle wrapper, device tests and Java formatting, the launcher icon and build configuration, then emulator-driven refinements and the final evidence and documentation. Describe the actual dates shown on GitHub accurately.

Live app demonstration 1 minute to 3 minutes 20 seconds

1:00 to 1:25 Open Pantry. Say: The pantry starts with stored ingredients, and each row shows its quantity and unit. Tap Add ingredient and briefly try saving without a name to show validation. Then enter bread with quantity 2 and unit piece, and save. This demonstrates Create and Read.

1:25 to 1:55 Add tomatoes 1 piece and olive oil 5 ml. Open Recipes and show Tomato toast. Say: The plural name tomatoes is normalized to tomato. Every ingredient has enough stock, so this recipe qualifies. Open the detail page and point to all three quantities and the numbered method.

1:55 to 2:20 Return to Pantry, edit bread from 2 to 1, save, and open Recipes. Say: The recipe disappears even though all ingredient names exist, because there is not enough bread. Change bread back to 2 and confirm the recipe reappears. This demonstrates Update and the quantity boundary.

2:20 to 2:45 Edit olive oil and choose Delete. Show the confirmation, then confirm. Reopen Recipes. Say: One missing ingredient is enough to exclude the entire recipe. There are no partial matches in this list. This demonstrates Delete and the central strict-matching rule.

2:45 to 3:20 Open Settings and toggle expiry badges. Return to Pantry. Force-stop the app through Android app settings, reopen it, and show that bread and tomatoes remain while deleted oil stays deleted. Say: SQLite persists the pantry beyond the Activity and process lifecycle. Settings also retain the chosen badge preference.

Strict matching 3 minutes 20 seconds to 4 minutes 15 seconds

Open app/src/main/java/za/ac/pantry/Matcher.java and cover lines 10 to 73.

> Explain canonical names and the mass, volume and count dimensions. Then show matches to the end of the file. Stock and requirements are aggregated in maps, expired stock is skipped, and any shortage immediately returns false. Extra pantry ingredients are harmless. Oil and water are never implicitly assumed.

Narration cue: describe what enters the method, how it is processed and what the caller receives. Point to the actual lines as you explain. Avoid reading every line aloud.

SQLite storage 4 minutes 15 seconds to 5 minutes 5 seconds

Open app/src/main/java/za/ac/pantry/Database.java and cover lines 40 to 70.

> Explain ContentValues, insertOrThrow, the parameterized update and delete selection, and pantry reads. Show the constructor and onCreate briefly as context: one local database stores pantry, recipe and requirement tables, with 20 recipes seeded once. Closing and reopening retains rows.

Narration cue: describe what enters the method, how it is processed and what the caller receives. Point to the actual lines as you explain. Avoid reading every line aloud.

Lifecycle and list display 5 minutes 5 seconds to 5 minutes 50 seconds

Open app/src/main/java/za/ac/pantry/PantryActivity.java and cover lines 10 to 67.

> Explain that returning from editing triggers onResume and reloads SQLite data. Then point to the PantryAdapter construction in render. The adapter supplies row count, stable database IDs and a card view for each ingredient. Intent extras carry IDs into the editor. BaseActivity performs database operations on a background executor.

Narration cue: describe what enters the method, how it is processed and what the caller receives. Point to the actual lines as you explain. Avoid reading every line aloud.

Database justification and close 5 minutes 50 seconds to 6 minutes 40 seconds

> Say: I chose SQLite because this app manages one household pantry on one device. It works offline, persists structured data, supports full CRUD and needs no account, remote server or API key. Recipe requirements use a foreign key to their recipe. SharedPreferences is sufficient for the small expiry badge setting. A cloud database would become useful if households needed to share synchronized stock.

Finish by showing the 17 matcher checks or Android validation output if time permits. Say: The tests cover missing ingredients, insufficient quantity, plural names, unit conversions, duplicate stock and expiry. The current app does not deduct stock automatically after cooking. A future improvement would add serving adjustment and a confirmed cook action.

Final recording checks: duration between 5 and 7 minutes, voice throughout, legible code and app screen, GitHub history shown, CRUD cycle completed, missing and insufficient ingredient behaviour demonstrated, process relaunch shown and three concepts explained. Export MP4 using H.264, place it directly inside the submission ZIP, and check the resulting file remains below 50 MB.